Scalability & Future Plan

Date	03 June 2026
Team ID	XXXXXX
Project Name	Raising waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The system uses historical weather data and does not support live weather updates.	Reduces real-time prediction capability.	High
2	The application is currently tested on a local Flask server.	Limited accessibility for multiple users.	Medium
3	The model is trained on a limited dataset.	Prediction accuracy may vary for different geographical regions.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users.	Deploy the application on IBM Cloud to support multiple concurrent users.
2	Data Storage	Uses a CSV dataset and local model files.	Integrate a cloud database to store weather data and prediction history.
3	Performance	Predictions are generated on a local machine.	Optimize the ML model and deploy it on cloud infrastructure for faster response times.
4	Security	Basic input validation is implemented.	Add user authentication, secure APIs, and encrypted communication (HTTPS).

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 1	Deploy the application on IBM Cloud.	Short Term	Improve accessibility and support online usage.
Phase 2	Integrate real-time weather APIs for live data collection.	Medium Term	Increase prediction accuracy and provide timely flood warnings.
Phase 3	Add SMS and Email alert notifications for high flood-risk predictions.	Medium Term	Enable faster communication with authorities and the public.
Phase 4	Develop a mobile application with GIS-based flood visualization and multilingual support.	Long Term	Improve usability, accessibility, and disaster response across different regions.